

Compare Sets by Counting On

Answer Key

Kindergarten–Grade 1

Quick Answers

1. Row A count = 4, Row B count = 6, which group has more (A, since $4 < 6$) = < 6. 3
 2. Row A count = 7, Row B count = 5, which row has fewer (B, since $7 > 5$) = > 7. 4
 3. Row A count = 6, Row B count = 6, more, fewer, or the same = = 8. Array A count = 12, Row B count = 9, which group has more (A, since $12 > 9$) = > 9. 4
 4. 5 10. Row A count = 7, Row B count = 9, how many more in Row B = 2, —
 5. Array A count = 9, Group B count = 7, which
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

The author marked 5 blanks as working space (*not* answers — nothing checks what lands there).

Curriculum

Level: Kindergarten–Grade 1

K.CC.C.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 23 tagged checks.

1. Row A (4 dots) and Row B (6 dots): count each, then which has more.

Solution: Count Row A one-to-one:

Count Row B:

When counting, 4 is said before 6, so 4 is less — Row **B** has more.

4
6
$4 < 6$

2. Row A (7 dots) and Row B (5 dots): count each, then which has fewer.

Solution: Row A:

Row B:

5 is said before 7 when counting, so Row **B** has fewer.

7
5
$7 > 5$

3. Two rows arranged differently: count each, then more, fewer, or the same.

Solution: Row A:

Row B:

Spreading dots out does not change how many there are.

6
6

The counts match, so the rows have the **same** number.

$$6 = 6$$

4. Row A has 3, Row B has 8: count on to find how many more.

Solution: Start at 3 and count on to 8: “4, 5, 6, 7, 8” — those are the hop lines. That is five hops, so Row B has more dots.

$$5$$

5. Array A (3 rows of 3) and Group B (7 dots): which has more.

Solution: Array A row by row: $3 + 3 + 3$ gives

$$9$$

Group B:

$$7$$

9 comes after 7 in the count, so Array **A** has more.

$$9 > 7$$

6. Ana has 4 grapes, Ben has 7: how many more does Ana need to have the same?

Solution: Count on from 4 to 7: “5, 6, 7” — three hops. Ana needs more grapes.

$$3$$

7. Ten-frames showing 6 and 10: how many more in the full frame?

Solution: Count on from 6 to 10: “7, 8, 9, 10” — four hops. The full frame has more dots.

$$4$$

8. Array A (3 rows of 4) and Row B (9 dots): which has more.

Solution: Array A row by row: $4 + 4 + 4$ gives

$$12$$

Row B:

$$9$$

12 comes after 9 in the count, so Array **A** has more.

$$12 > 9$$

9. Kim has 5 crayons, Jo has 9: how many more does Kim need?

Solution: Count on from 5 to 9: “6, 7, 8, 9” — four hops. Kim needs more crayons.

$$4$$

10. Challenge: Row A (7 dots), Row B (9 dots) — counts, how many more, and how you counted on.

Solution: Row A:

$$7$$

Row B:

$$9$$

Count on from 7 to 9: “8, 9” — two hops, so Row B has more dots.

$$2$$

Say how you counted on (instructor-judged): accept any wording that shows counting on from the smaller number (“I started at 7 and said 8, 9 — two hops”), not recounting both rows from one. Recounting still earns credit for the number if it lands on 2, but the strategy asked for is counting on.